Niklas Rieken

Curriculum Vitae

Komphausbadstraße 25
52062 Aachen, Germany
+49 1523 8736069
rieken@oms.rwth-aachen.de
ttnick.github.io

Personal Data

Date of Birth — November 9, 1992
Place of Birth — Krefeld, Germany

Education

May 2019 — **Master of Science**, *RWTH Aachen University*, Computer Science
Thesis: Network Pricing Games with Atomic Splittable Followers
Final Grade: 1.4

March 2017 — **Bachelor of Science**, *RWTH Aachen University*, Computer Science
Thesis: Alphabets and Machines as Structures and Their Decision Problems
Final Grade: 2.5

May 2012 — **Abitur (German A-Levels)**, *Michael-Ende-Gymnasium*, Tönisvorst
Advanced Courses: Math and Physics
Final Grade: 2.6

Work Experience

2019–today — **PhD Student**, *Chair of Management Science, RWTH Aachen University*
Researcher in Combinatorial Optimization and Algorithmic Game Theory.
Matchings, matroids, mechanism design, teaching courses in mathematical economics.

2017–2019 — **Research Assistant**, *Chair of Operations Research, RWTH Aachen University*
Developer for `GCG` (*generic column generation*, MIP solver).
Performance tests, extracting statistics, and collecting information on vast instance sets.

2014–2019 — **Teaching Assistant**, *Chair of Computer Science 7 / TCS, RWTH Aachen University*
TA for undergraduate courses in theoretical computer science.
Teaching exercise groups, grading assignments and exams.

Scientific Publications & Preprints

ANRW 2025 — **Using Explicit (Host-to-Host) Flow Measurements for Network Tomography**
with Ike Kunze, Constantin Sander, Alexander Ruhrmann, and Klaus Wehrle

TEAC 2025 — **Faster Dynamic Auctions via Polymatroid Sum**
with Katharina Eickhoff, Meike Neuwohner, Britta Peis, Laura Vargas Koch, and László A. Végh

Networks 2024 — **A flow-based ascending auction to compute buyer-optimal Walrasian prices**
with Katharina Eickhoff, S. Thomas McCormick, Britta Peis, and Laura Vargas Koch

WINE 2023 — **Faster Ascending Auctions via Polymatroid Sum**
with Katharina Eickhoff, Britta Peis, Laura Vargas Koch, and László A. Végh

MAPSP 2022 — **A Primal-Dual and Primal-Greedy Approximation Framework for Weighted Covering Problems**
with Britta Peis, José Verschae, and Andreas Wierz

(Preprint) — **A Simplified Analysis of the Ascending Auction to Sell a Matroid Base**
with Britta Peis

Scientific Talks

July 3, 2025 — **A Self-Reflecting Greedy Algorithm for Submodular Cover**
10th Gerhard Woeginger Research Colloquium, RWTH Aachen University

May 27, 2025 — **Combinatorial Optimization in Auction Design**
Seminar on Microeconomics, RWTH Aachen University

November 28, 2023	**Selling Bases of a Matroid** ORM PhD Seminar, RWTH Aachen University
August 30, 2023	**An Ascending Auction for Computing Minimal Walrasian prices via the Matroid Partitioning Algorithm** OR 2023, Universität Hamburg
June 15, 2023	**Selling Bases of a Matroid** 13th Day on Computational Game Theory, Vrije Universiteit Amsterdam
January 18, 2023	**Computing Walrasian Prices via Matroid Partitioning** Santiago Workshop on Combinatorial Optimization, Universidad de Chile
July 6, 2022	**Computing Buyer-Optimal Walrasian Prices in Multi-Unit Matching Markets via a Sequence of Max-Flow Computations** Seminario AGCO, Universidad de Chile

Awards

November 9, 2024	**Prize for outstanding teaching activities** Faculty 8 of Economics, RWTH Aachen University

Coursework

- Complexity Theory
- Recursion Theory
- Mathematical Logic I + II
- Logic and Games
- Applied Automata Theory
- Infinite Computations
- Convex Optimization
- Operations Research I + II
- Optimization with Modelling Languages
- Methods and Applications of Optimization
- Discrete Differential Geometry
- Satisfiability Checking

Skills

Operating Systems	**GNU/Linux**
Programming	**Python, Bash, C/C++**
Applications	**LaTeX(+Ti*k*Z), Gurobi**

Languages

German	native
English	proficient
Spanish	basics

Professional Interests

- Mathematical Optimization
- Theory of Formal Languages
- Algorithmic Game Theory

Personal Interests

- Ultimate and Table Tennis
- Guitar and Ukulele
- Reading, Movies, and Music
- Cooking, Board games

Niklas Rieken
Aachen, July 30, 2025